

# AI-Driven Phishing Email Detection

Using Natural Language Processing and Machine Learning



# The Phishing Problem

Phishing is a type of cybercrime in which attackers send fraudulent emails that appear to originate from trusted organizations. The primary objective is to deceive users into revealing sensitive information such as usernames, passwords, and banking credentials.

## Rule-Based Filters Fall Short

Traditional spam filters fail to detect sophisticated, evolving phishing attacks.

## Attackers Adapt Constantly

Email content, sender addresses, and writing patterns are continuously modified to bypass defenses.



# Why Traditional Systems Fail



Traditional spam filters mainly rely on predefined rules and blacklists to identify malicious emails. These methods often fail to detect newly created phishing attacks because attackers continuously modify email content, sender addresses, and writing patterns.

## Low Detection Accuracy

New phishing variants easily slip past static rule sets.

## High False Positive & Negative Rates

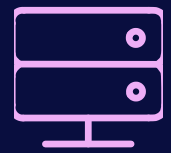
Legitimate emails blocked; malicious ones delivered.

## Cannot Adapt

No learning capability to evolve with new attack techniques.

# Technology Stack

The system integrates modern web technologies with machine learning libraries to provide comprehensive phishing detection. The architecture separates frontend, backend, and ML components for scalability and maintainability.



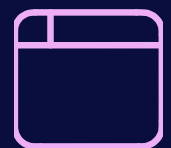
## Backend

Python & Flask power the application logic and REST API endpoints.



## Database

MySQL stores user data, email submissions, and prediction history.



## Frontend

HTML5, CSS3, Bootstrap 5, and JavaScript deliver a responsive UI.



## ML Libraries

Scikit-learn, NLTK, Pandas, and NumPy drive model training and analysis.

# System Architecture & Features



The application enables users to register securely, submit email details for analysis, and review prediction history through an intuitive web interface. The system maintains a comprehensive database of analyzed emails and provides real-time performance metrics.

- Secure user registration and authentication
- NLP-powered email preprocessing and metadata extraction
- Multiple ML algorithms compared for optimal performance
- Interactive dashboard with prediction history and visualizations



# Machine Learning Approach

The system preprocesses email subject and body using Natural Language Processing techniques and converts textual content into numerical features using **TF-IDF (Term Frequency–Inverse Document Frequency)**. Multiple machine learning algorithms are trained and evaluated to identify the best-performing model.

## TF-IDF Feature Extraction

Transforms raw email text into high-dimensional numerical vectors that capture the statistical significance of each term across the corpus.

## NLP Preprocessing Pipeline

Tokenization, stop-word removal, and stemming prepare raw text before vectorization.

## Models Evaluated

01

### Logistic Regression

Best performer — F1 Score: 0.8

02

### Naive Bayes

Probabilistic baseline classifier.

03

### Random Forest

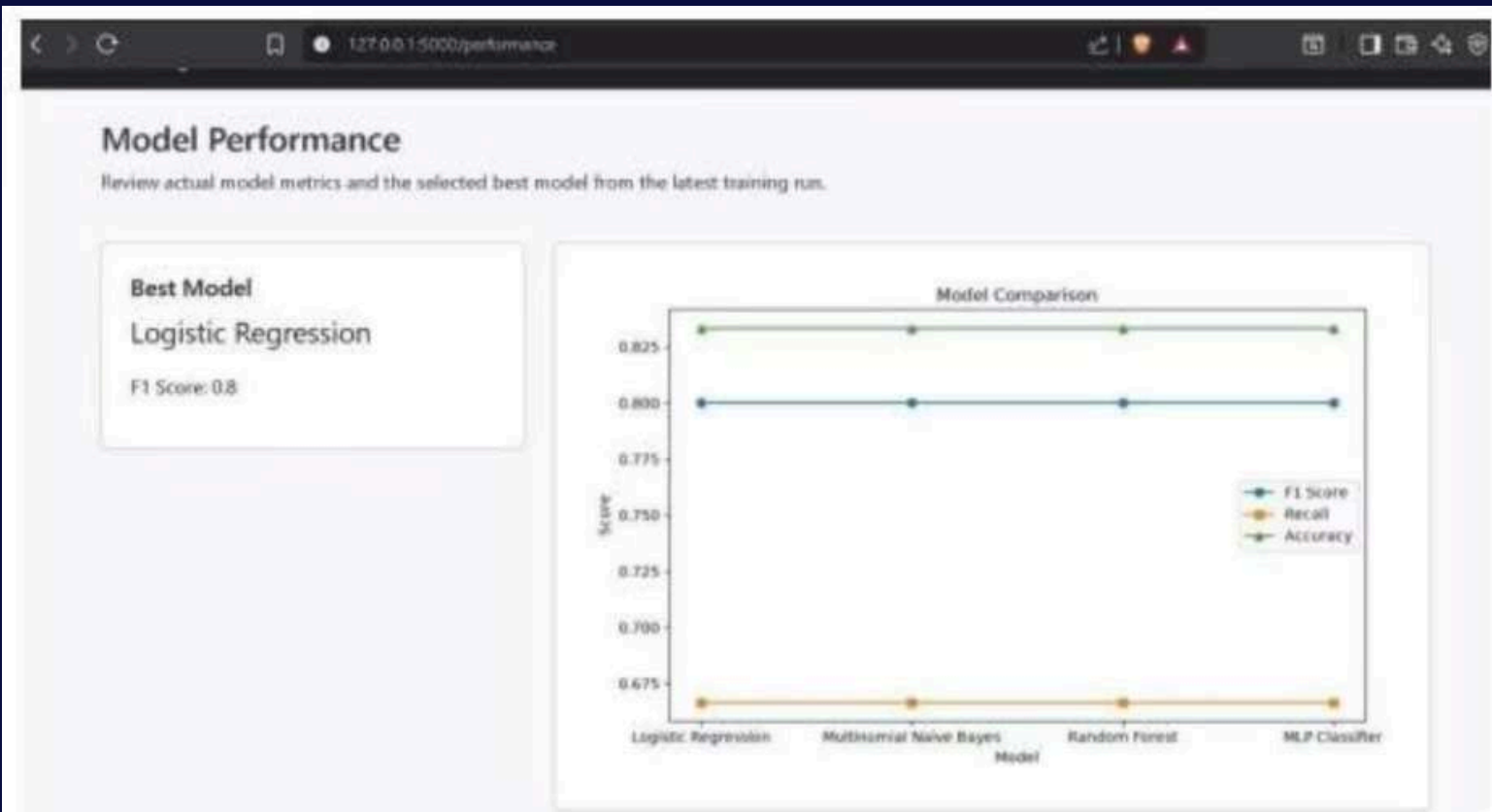
Ensemble tree-based approach.

04

### MLP Classifier

Multi-layer perceptron neural network.

# Model Performance Comparison



Logistic Regression outperforms all other models across F1 Score, Recall, and Accuracy metrics — making it the selected classification model for deployment.

0.8

F1 Score

Logistic Regression

4

Models Tested

LR, NB, RF, MLP

Logistic Regression was selected as the production model based on its balanced precision and recall performance.



# Key Advantages & Future Scope

The proposed system provides an efficient, accurate, and user-friendly approach to phishing email detection by combining intelligent text analysis with machine learning. It enhances cybersecurity awareness and establishes a scalable foundation for future enhancements.

## ✓ Accurate ML Detection

NLP-powered analysis delivers high-precision phishing classification in real time.

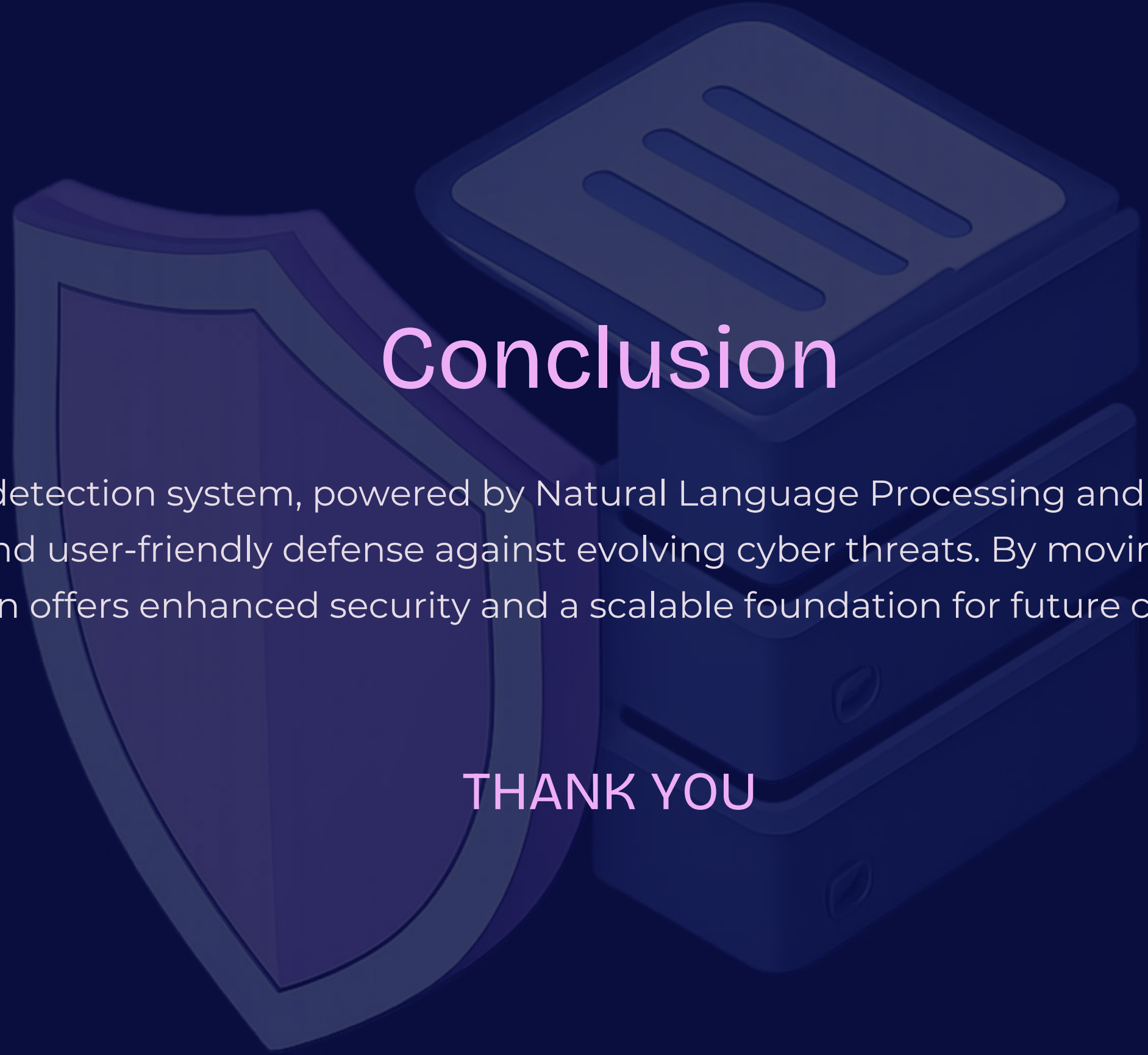
## 🔒 Secure & User-Friendly

Authenticated access with an intuitive dashboard for all user types.

## 🚀 Future Roadmap

Real-time email integration, deep learning models, and cloud deployment are next.



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# Conclusion

Our AI-driven phishing detection system, powered by Natural Language Processing and Machine Learning, provides an accurate, adaptive, and user-friendly defense against evolving cyber threats. By moving beyond traditional rule-based filters, this solution offers enhanced security and a scalable foundation for future cybersecurity advancements.

THANK YOU